

Qn 6	Answer	Marks
(a)	The activity of a radioactive source is the <u>rate at which a source of unstable nuclei decays</u> or <u>the number of disintegrations per unit time.</u>	A1
(b)(i)	$N = (2.40 \times 10^{-6}) \left(\frac{6.02 \times 10^{23}}{235} \right)$ $= 6.148 \times 10^{15}$	M1 A1
(b)(ii)	Decay constant $\lambda = \frac{A}{N} = \frac{0.1919}{6.148 \times 10^{15}}$ $= 3.1213 \times 10^{-17} = 3.12 \times 10^{-17} \text{ s}^{-1} \text{ (to 3 s.f.)}$	M1 A1
(c)(i)	${}_{19}^{42}\text{K} \rightarrow {}_{20}^{42}\text{Ca} + {}_{-1}^0\text{e} + \text{antineutrino}$ beta particles are emitted. Accept electrons. Accept: beta with (anti-)neutrino	A1
(c)(ii)	Half-life = 12.5 h	A1
(c)(iii)	Correct Shape B1 sum of Calcium-42 and Potassium-42 equals 1.0 N ₀ at $t = 0$, $t = t_{1/2}$ and $t = 40 \text{ h}$	B1 B1
(c)(iv)	Four parts Calcium-42 and one part Potassium-42. From the graph, we get 0.8 N ₀ for Calcium-42 and 0.2 N ₀ for Potassium-42 giving the age as 29 hours. <u>Alternatively,</u> Ratio of Calcium-42 to Potassium is 4:1 Hence, for K-42, $N = N_0 e^{-\lambda t} \Rightarrow t = -\frac{t_{1/2}}{\ln 2} \ln\left(\frac{N}{N_0}\right) = -\frac{12.5 \text{ h}}{\ln 2} \ln\left(\frac{1}{5}\right)$ $= 29.0 \text{ h}$	M1 A1
[Total : 11 marks]		